

Delhi Public School

Subject: Biology | Grade: 12th | Class: 12th

Time: 2 Hours

Maximum Marks: 13

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

Section A

Objective Type Questions. Each question carries 1 mark. Choose the correct option.

Q1. Which of the following crosses is used by geneticists to determine the homozygous or heterozygous dominance of a trait in an F1 individual? [Easy] [1]

- a. Self-pollination of F1 generation
- b. Crossing F1 hybrid with the dominant parent
- c. Crossing F1 hybrid with the double recessive parent
- d. Crossing two heterozygous dominant parents

Expected Answer: Crossing F1 hybrid with the double recessive parent

Q2. The AB blood group in humans is a classic example of codominance. This means that in a heterozygous individual with IA IB genotype: [Easy] [1]

- a. Allele IA is partially dominant over IB
- b. Both alleles IA and IB express themselves fully and independently
- c. Allele IB completely masks the expression of IA
- d. A completely new intermediate antigen is formed on RBCs

Expected Answer: Both alleles IA and IB express themselves fully and independently

Q3. In a pea plant, a dihybrid cross is performed between a plant heterozygous for yellow and round seeds (RrYy) and another plant with green and wrinkled seeds (rryy). What is the expected phenotypic ratio of the offspring? [Moderate] [1]

- a. 9:3:3:1
- b. 1:1:1:1
- c. 3:1
- d. 1:2:1

Expected Answer: 1:1:1:1

Q4. According to the Hardy-Weinberg principle, if the frequency of a recessive allele 'a' in a stable population is 0.3, what is the expected frequency of the heterozygous genotype 'Aa'? [Moderate] [1]

- a. 0.09
- b. 0.49
- c. 0.42
- d. 0.21

Expected Answer: 0.42

Q5. Industrial melanism in the peppered moth (Biston betularia) in England during the industrial revolution represents a shift from light-colored moths to dark-colored moths. This phenomenon illustrates which type of natural selection? [Hard] [1]

- a. Stabilizing selection
- b. Directional selection
- c. Disruptive selection
- d. Artificial selection

Expected Answer: Directional selection

Section B

Short Answer Type Questions. Each question carries 2 marks. Answer in 30-50 words.

Q6. State Mendel's Law of Segregation and explain why it is also known as the 'Law of Purity of Gametes'. [2]

[Easy]

Expected Answer: Mendel's Law of Segregation states that during gamete formation, the two alleles of a gene separate from each other so that each gamete receives only one allele. It is called the 'Law of Purity of Gametes' because the alleles do not show any blending and gametes always remain pure for a particular trait.

Q7. Using a Punnett square, determine the phenotypic and genotypic ratios of the F₂ generation in a cross showing incomplete dominance, such as flower color in Snapdragon (*Antirrhinum majus*) crossed between red (RR) and white (rr) flowered plants. [2]

[Moderate]

Expected Answer: The F₁ generation produces pink flowers (Rr). Selfing F₁ (Rr x Rr) gives F₂: Genotypes: 1 RR (Red) : 2 Rr (Pink) : 1 rr (White). Thus, both Genotypic ratio and Phenotypic ratio are identical, i.e., 1:2:1.

Q8. Differentiate between homologous and analogous organs. How do they provide evidence for divergent and convergent evolution respectively? [2]

[Moderate]

Expected Answer: Homologous organs have the same basic anatomical structure but perform different functions (e.g., flippers of whales and wings of bats), proving divergent evolution from a common ancestor. Analogous organs have different origins but similar functions (e.g., wings of insects and birds), proving convergent evolution due to adaptation to similar ecological niches.

Section C

Long Answer Type Questions. Each question carries 1 mark. Answer in detail.

Q9. State the Hardy-Weinberg principle. Identify and evaluate any three major evolutionary factors that can upset the genetic equilibrium of a population, leading to evolutionary change. [1]

[Moderate]

Expected Answer: The Hardy-Weinberg principle states that allele frequencies in a stable population remain constant from generation to generation in the absence of evolutionary forces ($p^2 + 2pq + q^2 = 1$). Factors that upset this equilibrium include: 1. Gene migration/flow: Transfer of alleles into or out of a population changing allele frequencies. 2. Genetic drift: Random fluctuation of allele frequencies by chance, especially in small populations. 3. Natural selection: Differential survival and reproduction of genotypes, favoring beneficial traits and shifting gene frequencies over generations.

Q10. Imagine you are a plant geneticist. You cross a pure-breeding plant with blue flowers with another pure-breeding plant of the same species with yellow flowers. All the resulting F₁ plants produce green flowers. Formulate a hypothesis to explain this inheritance pattern. Outline a complete genetic cross (including parental, F₁, and expected F₂ phenotypic and genotypic ratios) to support your hypothesis, and design a test cross to confirm the genotype of the F₁ green-flowered plants. [1]

[Hard]

Expected Answer: Hypothesis: The inheritance pattern is Incomplete Dominance, where the green color is an intermediate heterozygous phenotype (C^BC^Y) resulting from the partial dominance/co-expression of blue (C^B) and yellow (C^Y) alleles. Genetic Cross: - Parents: Blue (C^BC^B) x Yellow (C^YC^Y) - F₁: All Green (C^BC^Y) - F₂ (F₁ self-pollinated): Genotypic ratio = 1 C^BC^B : 2 C^BC^Y : 1 C^YC^Y. Phenotypic ratio = 1 Blue : 2 Green : 1 Yellow (Ratio 1:2:1). Test Cross Design: Cross the F₁ green plant (C^BC^Y) with a homozygous parental strain, e.g., yellow (C^YC^Y). Expected Test Cross Result: 50% Green (C^BC^Y) and 50% Yellow (C^YC^Y) (1:1 ratio). This 1:1 ratio will confirm the heterozygous status of the F₁ green plant.

— End of Question Paper —